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Identifying Opportunities

Although most producer companies devote major resources to researching the market for their products, many products still appear on the market that seem not to have been designed with their user in mind. As modern products become more and more complex, with more and more functions and features available through their microprocessor components, many of them have become increasingly confusing to their users. At the same time, the availability of a greater diversity of products has led to greater consumer choice, and therefore the greater importance of consumer preferences. Consequently, a concern has grown to ensure that the design of products and systems becomes more user centred.

Of course, there is not usually just one user of a product. Sometimes the range of users of a product, and their different needs, can be very diverse. In addition to the obvious or intended users there is a variety of people who have to interact with the product in various ways at different times, such as the people who make it, assemble it, service it, and recycle it.

Many products do not work well in use because the presence of the user is not strongly represented during design. The designer is subject to many, often conflicting pressures that require trade-offs to be made between users' needs and other factors. For example, manufacturing requirements can often override users' needs. This is because manufacturing and assembly difficulties are evident to the producer, and cost money through lost production time, faulty products, accidents, and worker dissatisfaction.

A user-centred approach is often referred to as ‘universal’ or ‘inclusive’ design – meaning designing to include everyone. Inclusive design means designing products so that they can be used easily by as many people as wish to do so. This may sound an obvious goal, but the fact is that many people have difficulty carrying out ordinary tasks with everyday products. Many elderly or disabled people cannot carry out – certainly with any ease or dignity – the range of everyday tasks that others take for granted. They are forced to choose products from a limited range that may suit them, and may also have to adapt products themselves, or buy expensive attachments to compensate for a product’s inadequate design.

The *user scenarios* method provides a useful starting point and focus for the design process by taking the user’s point of view, and identifying opportunities for creating new products that aim at satisfying users’ needs.

The User Scenarios Method

Procedure

*Practice being
a user of a product
or service*

A simple first step in user research for design is to use a product or service yourself, but in an aware and critical frame of mind. This is called making a *user trip*, or user journey, in which you use the product or service in a deliberate way, and note down your reactions.

The essential idea of user trips is very simple: you just take a ‘trip’ through the whole process of using a particular product or system, making yourself a critically observant user. You may be surprised how much you find out, if you make yourself sufficiently self-aware and observant. In fact, after trying this method a few times, you will probably find that you are adopting an attitude to most of the products and systems used in everyday life, that other people might regard as hypercritical. But it is this critical attitude, this dissatisfaction with the accepted norms, that spurs designers to make improvements. The secret is to turn dissatisfaction into creative criticism, to develop an attitude of constructive discontent.

Decide first of all which user’s point of view you are taking: customer, operator, maintenance person, etc. You may want to make several trips, from different user perspectives, or try special user perspectives, such as that of a disabled person. It is usually easiest to take a customer’s trip, since you may need special permission, access, and perhaps skills, before you can take any other.

Next, decide the limits, and the variations, to the user trip or trips that you are going to take. It is usually a good idea to extend the trip into activities that both precede and succeed the immediate use of

the product you are investigating since this may lead you into devising an improved, more integrated overall solution. Similarly, variations on a basic trip (at different times of day, in different weather conditions, with different requirements) will probably bring to light a wider range of problems than would a single, random trip.

Now you just have to set out and take the trip for real, recording your actions, impressions, ideas and thoughts. On a user trip it is important to be on the lookout for even the very tiniest problems or discoveries, because they might add up to something significant. And don't ignore even your most trivial-seeming ideas for improvements, because again they might add up to something bigger, or lead later to an important insight into design faults. This recording of thoughts, impressions, etc., ideally should be done during the trip, or else as soon as possible afterwards. A pocket audio recorder is very useful; otherwise use a notebook.

If you are an experienced user of the product or system (using a familiar machine or following a regular procedure), remember that you will have adapted yourself to many of the difficulties involved, and will probably be unaware of the impressions that an inexperienced user would have. So try to make yourself again as naïve as you must once have been, and hypersensitive to even the slightest difficulties or impressions you may have. Alternatively select a situation or trip variation that makes you more like an inexperienced user, such as using a new piece of equipment, or following an unfamiliar routine.

Observe users in action

User trips can generate a surprisingly high number of ideas for improvements to everyday products. But user trips suffer from a number of shortcomings that limit their usefulness. For example, a single product can have many different kinds of users and it is often difficult to take the viewpoint of a user who has a role different from your own. Or you might make mistakes that an experienced user would not, thus obtaining a wrong impression of the product. One way around these difficulties is to observe other people performing as users and to choose different kinds of experienced users, depending on what information you want to gather.

Begin by identifying those *experienced users* who will be able to provide you with relevant information. Most people are usually very willing to cooperate if approached tactfully and with obvious good intent. Get them to describe and demonstrate what seems important to *them*, as well as on what *you* regard as important. Look for, and ask about, aspects that seem difficult or unusual or critical. Look also for informal modifications to the job, workspace or equipment that the

user may have made for themselves (e.g., a hand-made sign or label on a machine, additional loose cushions on a seat, etc.). Record what you see and are told, and your impressions, during or immediately after the observation.

An amazing amount of information relevant to the redesign of a product or system also can be gained from observing *inexperienced users'* attempts to cope. Inexperienced users can reveal potentially important problems, difficulties, ideas, etc., of which experienced users will be unaware.

Finding an inexperienced user is not always easy, but there may be ways of lowering a user's normal level of experience by introducing some novel feature. Acting as an inexperienced user, and perhaps being made to look naïve or foolish, can be embarrassing, too. You should reassure your volunteers (and maybe yourself), that you are looking for shortcomings in the product design, not in the users, and that they should blame the design of the product, not themselves, for any difficulties.

Once you have found volunteers, give them only the basic objective they have to achieve, without any detailed instructions. Such an objective might be 'set the central heating to switch off for two days next weekend', or 'listen to the voice-mail message that has been received on this telephone, and send a short text reply'.

You need to observe and record very carefully what the inexperienced user does. Get your volunteers to talk about the task as they attempt it. You may find that it takes what you would regard as an extraordinary amount of time for an inexperienced user successfully to operate something that you are familiar with yourself, and your volunteer may actually be unable to achieve a seemingly simple objective. Don't offer advice if your volunteers get stuck (unless it looks as though they will cause damage or an injury, or until it would be more fruitful to move on to the next stage of a sequence of operations), and try not to laugh or get angry. Remember the advice about blaming the design, not the user.

Question users about their experiences

All over the world, producer companies have increasingly learned to keep a careful watch on emerging consumer requirements and changing user needs and wishes. Techniques used in market research to gather consumers' views on products include both quantitative methods (collecting numerical data), such as survey questionnaires, and qualitative methods (collecting opinions and subjective responses), such as focus groups.

Most questionnaires require a fixed type of response, such as a choice between available answers, or along a scale of response. For

example, a product design questionnaire might suggest 'I found the product easy to use' and provide a 5-point scale of response from 'agree strongly' to 'disagree strongly'. Advantages of fixed-response types of questionnaires are that they are quick to complete, lend themselves to easy processing of the responses, and result in numerical data.

Another type of questionnaire asks for open-ended responses, with questions such as 'Are there any features of the product that you dislike?' or 'Would you recommend this product to a friend – if so, why?' Open-ended questionnaires can provide useful insights gleaned from the responses, or can provide data such as how often a product feature is mentioned in free responses. But they require more processing of the responses, and provide more qualitative feedback and less numerical data than fixed-response questionnaires.

A 'focus group' is simply a group of people gathered together to discuss a particular issue. In research for product design, a focus group might be a group of purchasers of a particular product brought together to discuss their feelings and attitudes towards the product and rival products; or perhaps their general likes and dislikes about those types of products.

A focus group will typically have between six to a dozen participants, plus a discussion leader (the researcher). The leader will have a list of issues to guide the discussion, but will allow the participants to develop their own topics of discussion, so as to let the discussion go in directions favoured by the participants. Examples of particular products to be discussed will usually be available to view or handle.

*Create relevant user
personas and
scenarios*

How can we draw together the varied needs and preferences, wishes and requirements into a 'model' of users of a new product, in order to provide some guidelines for the product designer? How can we use rather vague but inter-related sets of user preferences to identify the potential for new products to meet unsatisfied – and even unrecognised – user requirements? One way is to construct *scenarios* for how a product might be used by different users – that is, to construct some 'story lines' of the user's lifestyle and requirements that relate to opportunities for new product development.

Design scenario writing begins with identifying the potential *user profiles* and *personas*. A user profile summarises the characteristics of the user population that must be designed for. A user persona is a hypothetical, but well-defined, particular user within that population. Personas are not real people; rather, they are imaginary examples of real users they represent.

In defining personas you should be as specific as possible about the ‘made up’ details, and you should also give the persona a name, and possibly an image, whether it is an anonymous photograph or a sketched caricature. During the design process, the persona is referred to by name, rather than as ‘the user’. A set of personas can be defined for a project, something like a ‘cast of characters’ who might use the product. Every cast, though, should have at least one primary persona who is the main focus of your design.

Observing and talking to users can provide starting points for identifying profiles and personas. For instance, you can observe the range of different users of a particular product. Or if you know some people who are likely to be users of a certain product type, you can draw upon your personal knowledge of these people, or you can ask them to complete a questionnaire about their lifestyle, likes and dislikes. Different product types will have different user profiles.

Define the preliminary goal, context, constraints and criteria for a new product opportunity

The formal starting point for the design of a new product is the design brief. Although it is a very important step in the product planning and development process, it does not always get the attention it deserves. A design brief is the instruction to the designer from the client to take on a project, and some background information on the decision to initiate the project. However, in many examples of product design, this instruction is ‘brief’ in another sense – i.e. it contains very little information!

The design brief is an important step in the clarification of just what a new product is expected to be and to do. It is not, of course, a statement of the design *solution* – which is what the designer(s) will eventually generate – but a statement of the design *problem* or *opportunity*.

The design brief is an intermediate stage between the fundamental or underlying product idea and a full specification of the requirements that the product is expected to satisfy. (Developing a full specification is dealt with in the performance specification method, Chapter 8.) Crucially, the brief sets the design *goal*, outlines the *context* in which the new product must operate, identifies the major *constraints* within which the design goal must be achieved, and suggests some *criteria* by which a good design proposal might be recognised. The difference between constraints and criteria is that constraints set specific (usually quantitative) targets or limits for the product designer, whilst criteria are more flexible and might be used for judging between different product design proposals, each of which meets the specific constraint targets.

Summary

The aim of the user scenarios method is to identify and define an opportunity for a new or improved product. The procedure is as follows.

1. Practice being a user of a product or service.
Decide which user's (or users') point(s) of view to adopt and the variations to the user trip or trips you are going to take.
2. Observe users in action.
Both experienced and inexperienced users can provide valuable insights.
3. Question users about their experiences.
This can include the use of formal, structured or unstructured questionnaires, and focused group discussions.
4. Create relevant user personas and scenarios.
A persona is a well-defined but hypothetical user, and a scenario is a storyline about their use of the product or service.
5. Define the preliminary goal, context, constraints and criteria for a new product opportunity.
These are the key steps in formulating a good brief for a new product design.

Examples

Example 1: Sewing machine

A manufacturer of domestic sewing machines wanted to re-vamp one of their products. Their aim was to keep their product looking fresh, in order to maintain its market position. They asked designer Kenneth Grange to suggest a new design, perhaps looking only for some restyling. The resulting new product design incorporated the manufacturer's standard machinery, but repackaged it in novel ways that not only made it easier to use but also gave the machine a new and distinctive form and style.

The origins of the new design features lay in the designer's personal experience through a user trip with the existing model of sewing machine, using it for some simple sewing tasks. The designer found it strange that the sewing arm was located centrally over its base, whereas he realised that the user needs more surface area on their side of the needle than behind it. In front of the needle, space is needed to prepare the material for sewing; once the work has passed beyond the needle the user can do nothing more, it's sewn. He therefore proposed moving the sewing machine mechanism rearwards

on its base, creating an off-centre layout with more base-table area in front of the needle than behind it. Traditionally, the sewing machine had been designed as a very straightforward, basic piece of engineering with the mechanism placed centrally upon the base, and nobody had thought about challenging this.

Another radical change in this particular design was also a result of a simple assessment of how the machine is used. The designer gave the base of the machine rounded lower edges, which look like a mere styling feature, but in fact also arose from considering use. A frequent problem with sewing machines is that a lot of lint and loose fibres come off the fabric. This gets down into the bobbin and shuttle mechanism below the needle, may get picked up and sewn into the workpiece, and at worst may stop the machine altogether. The traditional way of dealing with this was to require the user to open the base to get at the mechanism for cleaning. This was usually achieved by tilting the machine away from the user, into an unstable position that only allowed restricted access to the mechanism.

The designer decided to make access to the underneath much easier, by tilting the machine lengthways. That action in itself suggested a shape by giving the lower side edge of the machine a rounded shape to make it easier for the user to tilt the machine, and the underside was more easily accessible for cleaning and oiling the lower mechanisms. A similar radiused top front edge was also provided to the base plate, to allow the fabric to slide over it more easily, and various other features were added, such as small drawers for holding accessories. Kenneth Grange's sketches (Figure 5.1) help explain these ideas and features.

Example 2: Digital radio

When a broadcasting company wanted to explore the potential of digital audio broadcasting as it emerged into public use in the early 2000s, it asked a design consultancy to suggest new digital radio product concepts. The team used personas and scenario building to help generate ideas for the functions, operation and appearance of new radios. The ideas included visual displays built into the radios to complement the audio programme, such as musician portraits, programme guides, etc. One project focused on special purpose radios used by one person in different situations. Figure 5.2 shows one user persona that the designers created – Jean, whose work sometimes requires her to be on 24-hour call. The comments attached to the scenarios include ideas for features to be incorporated in a new radio design, prompted by the scenarios. (Source: Myerson/IDEO)

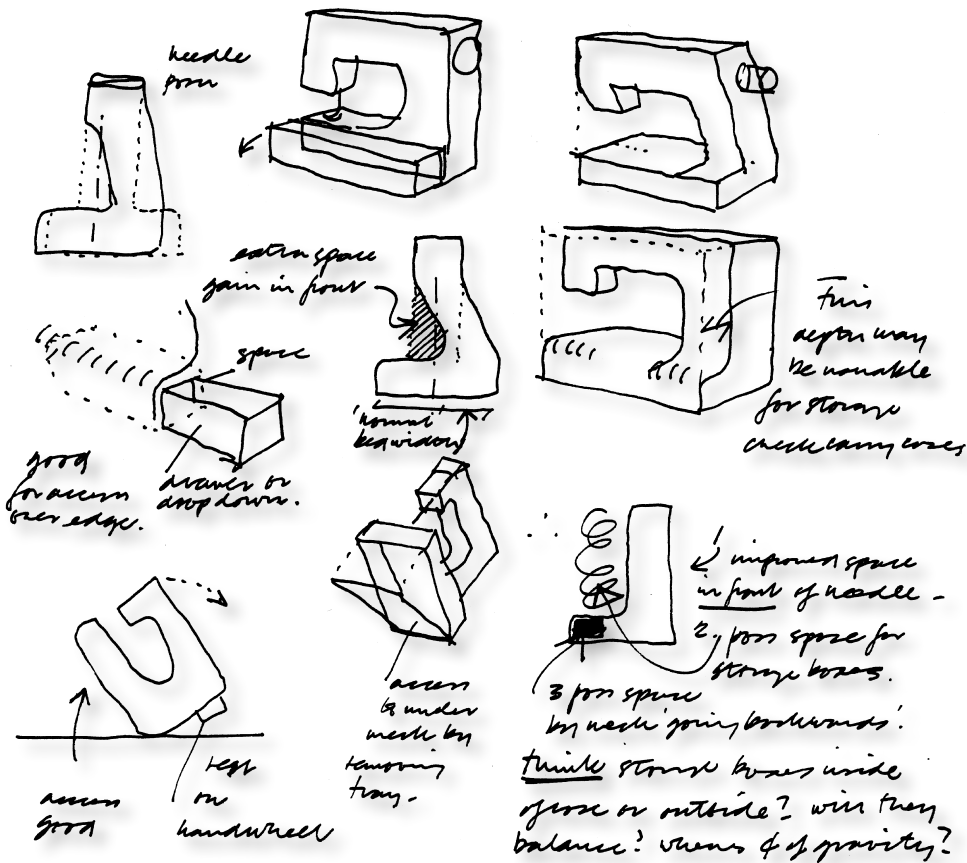


Figure 5.1 Design sketches for the user-centred redesign of a sewing machine. Source: Kenneth Grange.

Example 3: Oven aid

In this example, observation of elderly people using equipment in their kitchens led to identification of a new product opportunity. The designers developed a persona 'Mary', and identified a particular problem that she (and millions like her) has in using her oven. The example develops a new product opportunity from the persona and a user scenario.

Background:

Many elderly people experience difficulty with lifting things into and out of their ovens. This spoils their enjoyment of an everyday task, cooking, and restricts their independence in being able to perform such a task. The difficulties can be due to a variety of physical factors such as arthritis and rheumatism in hands, arms and back, muscular weakness, and limitations in movements such as bending and reaching.

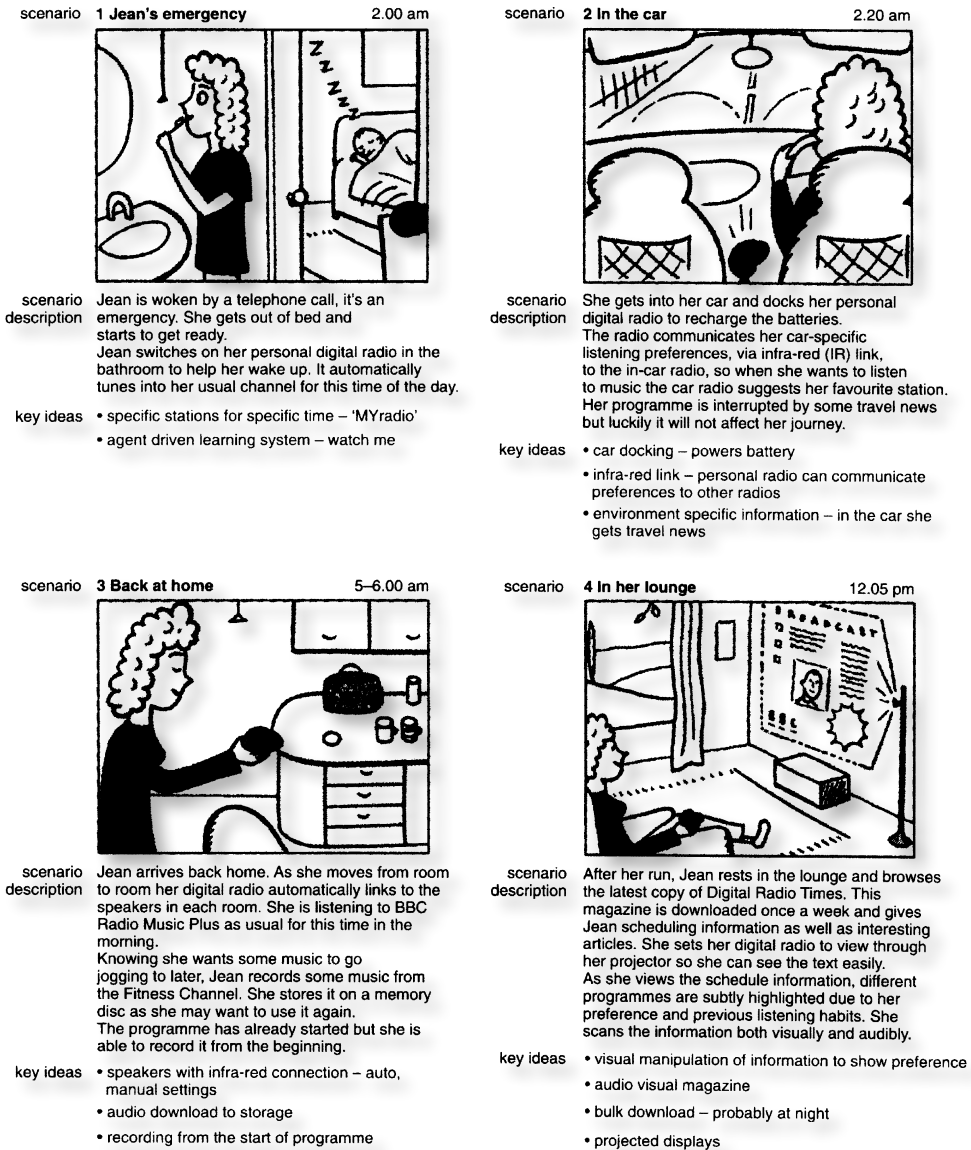


Figure 5.2 Scenarios for digital radio use by a shift worker. Source: Myerson, IDEO. Reproduced with permission of IDEO.

The primary product users in this context were considered to be elderly women. Although the problem is applicable to men as well, the majority of the elderly population is female. Any particular issues for women may make the product better meet the majority needs, wants, and desires of this population.

As well as observing elderly women performing cooking tasks with their ovens in their own homes, the design team consulted expert advisers such as health care workers who work with the elderly and doctors who work with seniors and are experts in rheumatology. They also researched guidelines for designing for disability. Other stakeholders consulted included people who install ovens and sell appliances, and organisations that promote products for seniors. It was also important to know about ovens, especially the type of ovens that older women might own.

Persona:

Mary is 70 years old and lives alone, but loves to bake and to entertain her family at home for holidays. Mary has arthritis in the lower spine and shoulders that limits her range of motion. She also has lost strength in her back and arm muscles. She is no longer comfortable reaching into the oven to lift things out. Losing the ability to bake things has been very depressing for her to contemplate. Mary is becoming hesitant to have her family over for meals and no longer feels confident entertaining in her home.

User scenario:

Mary needs help in lifting relatively heavy dishes into and out of her oven. A device is needed that fits with a standard oven and will compensate for Mary's limited motion and reduced strength and allow her to easily put in and remove from her oven a variety of pans and baking dishes. The device must have a simple, robust operating mechanism and must be inexpensive to buy and install. While the primary user market is senior women between the ages of 70 and 85, the primary purchasers may be family members. Any installation should be easy enough for a family member to do. The product must integrate with a range of standard ovens and be easy to clean.

Product Opportunity Proposal:

This product opportunity can be summarised in terms of the design goal, context, constraints and criteria as follows:

Goal:

An aid to assist lifting pots and dishes into and out of domestic ovens

Context:

Elderly people, especially women, who have problems lifting heavy dishes into and out of their ovens in everyday cooking

Constraints:

- usable by most women aged 70-85

- selling price not more than £30
- can lift and support a range of cooking pots, baking trays, etc. and their contents
- integrates with standard conventional ovens
- safe
- simple installation

Criteria:

- attractive to purchasers as well as users
- easy to use
- appears robust and safe

(Source: Cagan and Vogel)

*Worked example:
Message recorder*

An electronics communications company appointed a design consultant to identify areas for new product opportunities. They wanted suggestions for design briefs that could be handed to their own product designers for development into specific product concepts. One of their product ranges that they were concerned about was telephone answering machines, where sales were dropping.

Here is an outline of how the consultant approached that area, beginning with an exploration of the use of telephone answering machines. She realised that they are a product type that may have begun to outlive its natural life, as many users have taken up greater use of mobile phones with text and voice-mail facilities. However, there might still be opportunities for products of this type, aimed at existing markets, such as office telephones, or specific consumer markets, such as older people who only made limited use of mobile phones.

The consultant noticed that she and her family have two ageing answering machines in their house, but they often do not set them to record, or forget to check for messages. Also, she seems to be the only person who can operate them, being the only one who has read the manuals. Their first machine was bought about 12 years previously, and is integral with a fax machine. The second machine was bought about five years ago, and is integral with a telephone handset. Both machines seem to be confusing to inexperienced users, and inadequate for normal family use, so she thinks there is definitely scope for improvements.

The consultant first tried observing other users. She asked a variety of family members and friends to use the two answering machines. She gave them the same set of tasks for each machine, which were to:

- (a) listen to the previously received message that she had telephoned into both machines;
- (b) act on the message, which was to note or remember a telephone number and pass it on to someone else;
- (c) save the message;
- (d) record their own answer 'announcement', which will be heard when anyone phones in and gets the answer machine;
- (e) set the machine to receive messages.

The consultant asked these trial users to think aloud as they were doing these tasks, and recorded what they said on a small audio recorder. She also made her own notes as each trial user performed the tasks. For example, she noted that most users found it difficult to memorise the telephone number that the caller left in the message when there was no pen and paper at hand to write it down. From this observation, she noted the idea that an answering machine might be able to provide a printout of callers' numbers, with date and time of the call.

The consultant went on to conduct some focus group research. She asked small groups of target group users, aged in their 50s and 60s, to discuss answering machines together, using pairs of various machines as examples to initiate the discussion. As the group facilitator, she had prepared several prompts to keep the discussion going, and to move it into areas that she thought might be interesting. Some of the prompts were:

- What differences do you see between the two machines?
- What are the poor aspects of answering machines that you can recall from your own personal experiences?
- How do you feel about finding that your call is being taken by an answering machine?
- Can you recall either a funny event or a serious problem arising from missing a telephone message or misunderstanding a message?
- Besides telephone messages, what other messages do you leave around the house? Are there messages you'd like to leave, but cannot?

One especially interesting idea to emerge from the focus groups was a suggestion that an answering machine might be a more general

‘message machine’ that would allow anyone to record a message on it (a voice recorder) for anyone and everyone else in the household who comes home later – for example, ‘I’ve gone to the gym’ or ‘your dinner’s in the oven’. The consultant realised that her own family members often left written messages for one another on sticky notes on the kitchen wall or by the entrance door. She also realised that her own family household had diversified as children grew up, with the daily household activity patterns becoming more diverse as people came and left at different times. Therefore, a domestic messages recorder-player could be useful as a general purpose device, not just for telephone messages.

The consultant developed the following personae, scenario and product design opportunity proposal.

Personae:

Married couple Brian and Brenda are aged 55 and 53. Both are quite active, inside and outside the home, including work, sports and social activities. They have three children - two teenage sons at school and an adult working daughter, all living at home. Brenda’s parents, in their mid-to-late 70s, also play parts in their lives and home, sometimes coming into the house to help or to prepare meals.

User scenario:

Home life can be hectic and requires some co-ordination regarding meal times and preferences, visitors, schedules and appointments. The number of people at home at any one time is variable. Telephoning and messaging between each other can become complicated, and knowing and understanding separate plans and requirements can be difficult. Quite often written messages are left on the kitchen table, or on sticky notes by the entrance door, from individuals alerting some or all of the other family members to problems or plans. Brian and Brenda could make use of a domestic messages recording device that would allow them and all the other household members to leave individual or group messages for each other. Messages might be intended as one-to-one, or one-to-many.

Product opportunity:

There is a potential new market for domestic message recording and play-back devices for general use within a household. Such a device must be simple to use, across different age ranges. This implies easy-to-read graphics, and easy-to-use controls. The device must be clearly visible amongst household clutter (i.e. not easily lost like the TV remote control). It must be robust for frequent use by a

number of people. The device should be useable on a table top, and also potentially fixable in a permanent position, on a wall or a shelf. There should be a clearly visible alert signal that shows a message has been recorded. If feasible, message indicators might indicate which of several users left the message. As well as family use, variant devices might be marketed for use in other multi-person groups such as house sharing, or work groups sharing a communal space.

Summary proposal

Goal: A stand-alone message recording and play-back device.

Context: Domestic use by a range of family members or other communal house or work groups. Messages may be of various lengths, up to one minute. Different group members may listen and re-listen to messages at different times.

Constraints:

- usable by people of different age ranges, from 12 to 70+;
- simple, clear functions to record and play back messages;
- total recording time at least 15 minutes;
- messages lockable to prevent accidental erasure;
- clearly visible alert signal for messages stored;
- hand-held use or on table-top or fixed to wall or shelf;
- retail selling price under £30.

Criteria:

- easy to use;
- easily noticeable;
- user-friendly appearance.

6

Clarifying Objectives

When a client, sponsor or company manager first approaches a designer with a product need, it is unlikely that the 'need' will be expressed as clearly as in the previous chapter. The client perhaps knows only the type of product that is wanted, and has little idea of the details, or of the variants that might be possible. Or the 'need' might be much vaguer still: simply a 'problem' that needs a solution.

The starting point for design work is therefore very often an ill-defined problem, or a rather vague requirement. It will be quite rare for a designer to be given a complete and clear statement of design objectives. Yet the designer must have some objectives to work towards. The outcome of designing is a proposal for some means to achieve a desired end. That 'end' is the set of objectives that the designed object should meet.

An important first step in designing, therefore, is to try to clarify the design objectives. In fact, it is very helpful at all stages of designing to have a clear idea of the objectives, even though those objectives may change as the design work progresses. The initial and interim objectives may change, expand or contract, or be completely altered as the problem becomes better understood and as solution ideas develop.

So it is quite likely that both 'ends' and 'means' will change during the design process. But as an aid to controlling and managing the design process it is important to have, at all times, a statement of objectives that is as clear as possible. This statement should be in a form that is easily understood and that can be agreed by the client and the designer, or by the various members of the design team.

(It is surprising how often members of the same team can have different objectives!)

The *objectives tree* method offers a clear and useful format for such a statement of objectives. It shows the objectives and the general means for achieving them that are under consideration. It shows in a diagrammatic form the ways in which different objectives are related to each other, and the hierarchical pattern of objectives and sub-objectives. The procedure for arriving at an objectives tree helps to clarify the objectives and to reach agreement between clients, managers and members of the design team.

The Objectives Tree Method

Procedure

Prepare a list of design objectives

The 'brief' for a design problem is often very aptly called that – it is a very brief statement! Such brevity may be because the client is very uncertain about what is wanted, or it may be because they assume that the designer perfectly understands what is wanted. Another alternative is that the client wishes to leave the designer with as much freedom as possible. This might sound like a distinct advantage to the designer, but can lead to great frustration when the client decides that the final design proposal is definitely *not* what was wanted! In any case, the designer will almost certainly need to develop the initial brief into a clear statement of design objectives.

The design objectives might also be called client requirements, user needs or product purpose. Whatever they are called, they are the mixture of abstract and concrete aims that the design must try to satisfy or achieve.

Some design objectives will be contained within the design brief: others must be obtained by questioning the client, or by discussion in the design team. Typically, initial statements of objectives will be brief and rather vague, such as 'the product must be safe and reliable'. To produce more precise objectives, you will need to expand and to clarify such statements.

One way to begin to make vague statements more specific is, literally, to try to *specify* what it means. Ask 'what is *meant* by that statement'. For example, an objective for a machine tool that it must be 'safe', might be expanded to mean:

- low risk of injury to operator
- low risk of operator mistakes

- low risk of damage to work-piece or tool
- automatic cut-out on overload

This kind of list can be generated simply at random as you think about the objective, or in discussion within the design team. The client may also have to be asked to be more specific about objectives included in the design brief.

The types of questions that are useful in expanding and clarifying objectives are the simple ones of 'Why?' 'How?' and 'What?' For instance, ask 'Why do we want to achieve this objective?', 'How can we achieve it?' and 'What implicit objectives underlie the stated ones?' or 'What is the problem really about?'

Order the list into sets of higher-level and lower-level objectives

As you expand the list of objectives it should become clear that some are at higher levels of importance than others. Sub-objectives for meeting higher-level objectives may also emerge, and some of the statements will be means of achieving certain objectives. This is because some of the questions that you will have been asking about the general objectives imply a 'means-end' relationship – that is, a lower-level objective is a means to achieving a higher-level one.

An example is the statement 'automatic cut-out on overload' in the list above. This is not really an objective in itself, but a means of achieving an objective – in this case, the objective of 'low risk of damage to work-piece or tool'. In turn, this 'low-risk of damage' objective is itself a lower-level objective to that of the overall 'safety' objective.

Your expanded list of objectives will therefore inevitably contain statements at various levels of specificity. In order to clarify the various levels that are emerging, re-write your general list of objectives into ordered sets. That is, group the objectives into sets, each concerned with one highest-level objective. For example, one set might be to do with 'safety', another to do with 'reliability', and so on. Within each set, list the sub-objectives in hierarchical order, so that the lower level ones are clearly separated as means of achieving the higher-level ones. Thus, for instance, your 'safety' list might look like this:

Machine must be safe

Low risk of injury to operator

Low risk of operator mistakes

Low risk of damage to work-piece or tool

Automatic cut-out on overload

The list is now ordered into three hierarchical levels. It can sometimes be difficult to differentiate between levels of objectives, or different people in the design team may disagree about relative levels of importance of some objectives. However, exact precision of relative levels is not important, and you want only a few levels, about which most people can agree. For instance, in the above list, 'low risk of injury' might be considered more important than 'low risk of mistakes', but all three 'low risk' objectives can conveniently be grouped at about the same level.

The valuable aspect to sorting objectives roughly into levels is that it encourages you to think more clearly about the objectives, and about the relationships between means and ends. As you write out your lists in hierarchical levels, you will probably also continue to expand them, as you think of further means to meet sub-objectives to meet objectives, etc.

When you have quite a lot of statements of objectives, it makes it easier to sort them into ordered sets if each statement is written onto a separate slip of paper or small card. Then you can more easily shuffle them about into groups and levels.

Draw a diagrammatic tree of objectives, showing hierarchical relationships and interconnections

As you write out and shuffle your lists, you will probably realise that some of the sub-objectives relate to, or are means of achieving, more than one higher-level objective. For example, the sub-objective of 'low risk of damage to work-piece or tool' might be not only a means of achieving safety but also a means of achieving reliability.

So, a diagram of the hierarchical relationships of these few objectives and sub-objectives might look like Figure 6.1. This diagram is the beginnings of a 'tree' which shows the full pattern of relationships and interconnections. It is not necessarily just a simple 'tree' structure of branches, twigs, leaves, because some of the interconnections form loops or lattices. The 'tree' is also normally drawn 'upside-down' – i.e. usually it has increasingly more 'branches' at lower levels – and so it might be better to think of the sub-objectives as 'roots' rather than 'branches'.

It can sometimes be more convenient to draw the 'tree' on its side, i.e. with branches or roots spreading horizontally. In order to help organise the relationships and interconnections between objectives and sub-objectives, draw a complete 'tree' diagram, based on your ordered sets of objectives. Each connecting link that you

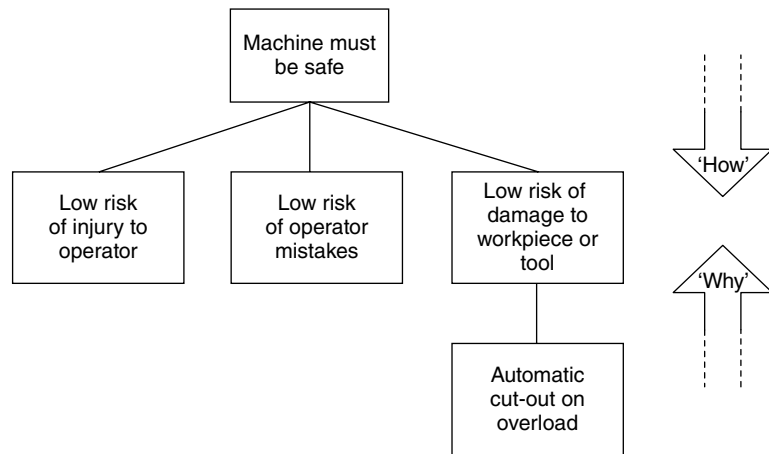


Figure 6.1 Hierarchical diagram of relationships between objectives.

draw indicates that a lower-level objective is *a means of achieving* the higher-level objective to which it is linked. Therefore working *down* the tree a link indicates *how* a higher-level objective might be achieved; working *up* the tree a link indicates *why* a lower-level objective is included.

Different people might well draw different objectives trees for the same problem, or even from the same set of objectives statements. The tree diagram simply represents one perception of the problem structure. The tree diagram helps to sharpen and improve your own perception of the problem, or to reach consensus about objectives in a team. It is also only a temporary pattern, which will probably change as the design process proceeds.

As with many other design methods, it is not so much the end product of the method (in this case, the tree diagram) which is itself of most value, but the process of working through the method. The objectives tree method forces you to ask questions about objectives, such as 'What does the client mean by X?' Such questions help to make the design objectives more explicit, and bring them into the open for discussion. Writing the lists and drawing the tree also begins the process of suggesting means of achieving the design objectives, and thus of beginning the process of devising potential design solutions.

Throughout a project, the design objectives should be stated as clearly as the available information permits; the objectives tree facilitates this.

Summary

The aim of the objectives tree method is to clarify design objectives and sub-objectives, and the relationships between them. The procedure is as follows.

1. Prepare a list of design objectives.
These are taken from the design brief, from questions to the client, and from discussion in the design team.
2. Order the list into sets of higher-level and lower-level objectives.
The expanded list of objectives and sub-objectives is grouped roughly into hierarchical levels.
3. Draw a diagrammatic tree of objectives, showing hierarchical relationships and interconnections.
The branches (or roots) in the tree represent relationships which suggest means of achieving objectives.

Examples

Example 1: City transport system

This is an example of expanding and clarifying design objectives from an initially vague brief. A city planning authority asked a transport design team for proposals for 'a modern system, such as a monorail, which would prevent traffic congestion in the city from getting any worse and preferably remove it altogether.'

The only clear objective in this statement is to 'prevent traffic congestion . . . from getting any worse. . .' But what are the implicit objectives behind the desire for 'a modern system, such as a monorail'? Traffic congestion might be held constant or reduced by other means.

By questioning their clients, the design team uncovered objectives such as a desire to generate prestige for the city and to reflect a progressive image for the city authority. There was also a wish simply to reduce complaints from citizens about the existing traffic system. It was also discovered that only certain types of new system would be eligible for a subsidy from central government.

The design team were able to draw up an expanded and hierarchically-ordered set of objectives, as shown in Figure 6.2. In particular, they identified a number of high-level, 'essential objectives' which were not explicitly stated in the original brief. By identifying these objectives, the designers clarified the project and the limitations that there might be on the range of alternative solutions. (Source: Jones)

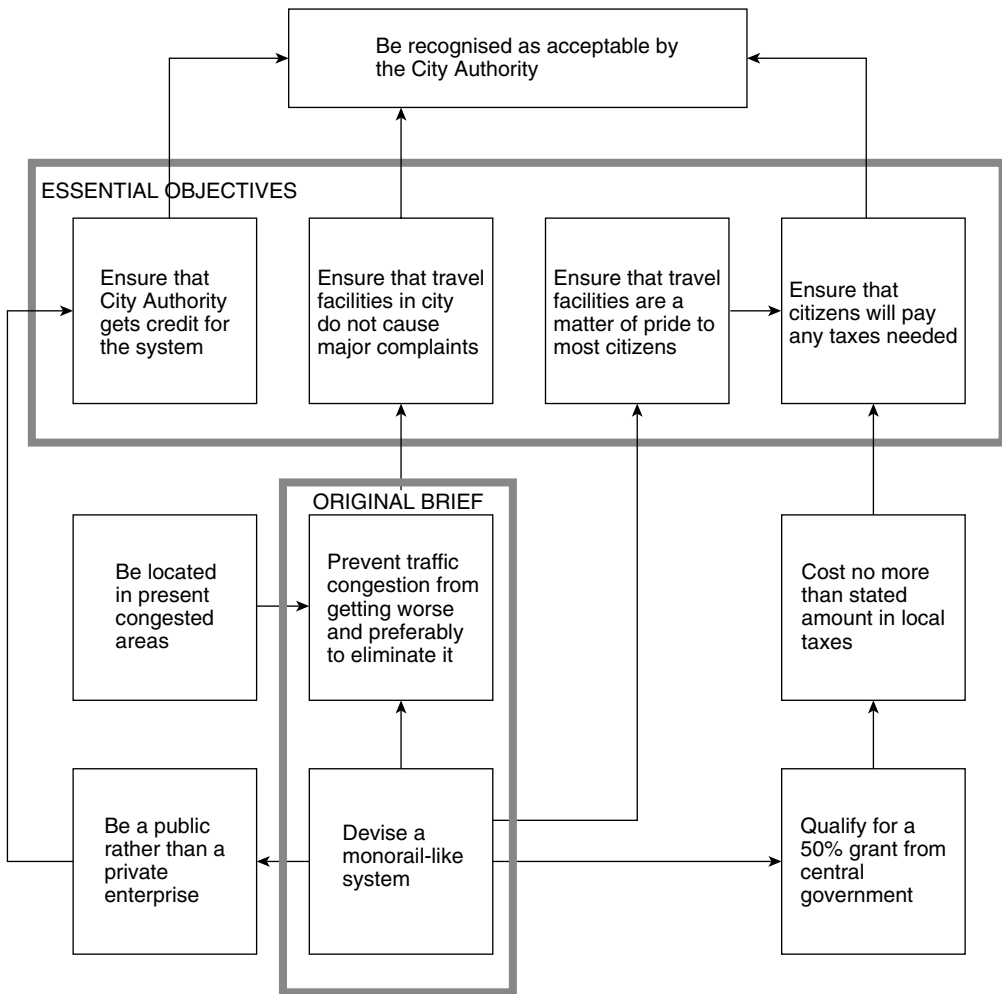


Figure 6.2 Expanded set of objectives for a new city transport system. Source: Jones, Design Methods. Reproduced with permission of Wiley.

*Example 2:
Impulse-loading
test rig*

An example of applying the objectives tree method in engineering design is provided here. The design problem was that of a machine to be used in testing shaft connections subjected to impulse loads.

As before, a typically vague requirement of a 'reliable and simple testing device' can be expanded into a much more detailed set of objectives (Figure 6.3). 'Reliability' is expanded into 'reliable operation' and 'high safety'. 'Simple' is expanded into 'simple production'

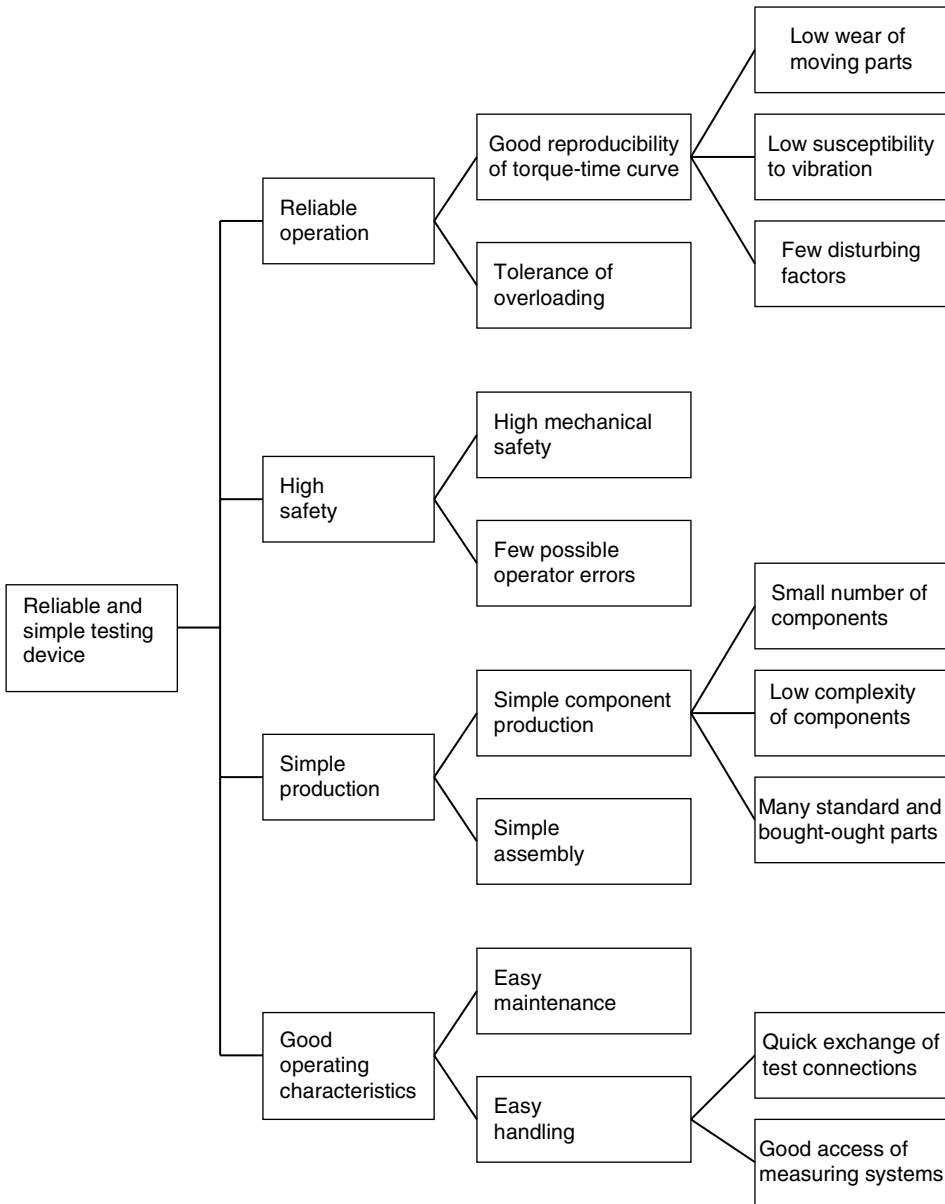


Figure 6.3 An objectives tree for an impulse-loading test rig. Source: Pahl & Beitz, Engineering Design. Reproduced with permission of The Design Council.

and 'good operating characteristics'; the latter is further defined as 'easy maintenance' and 'easy handling'; and so on.

In a case such as this, first attempts at expanding the list of objectives would probably produce statements at all levels of generality.

For example, asking 'What is meant by "simple"?' would have been likely to produce statements in random order such as, 'easy maintenance', 'small number of components', 'simple assembly', etc. Drawing these out in the hierarchical tree structure shows how they relate together. (Source: Pahl and Beitz)

*Example 3:
Automatic teamaker*

The objectives tree method can also be used in designing a relatively simple device such as an automatic teamaker. In this example, a distinction is made between 'functions' and 'means'. Each 'function' is an objective, which may be achieved by a number of different 'means' or sub-objectives. Thus the function 'combine water and tea leaves' could be achieved by adding the water to the tea, adding the tea to the water, or bringing them both together into one receptacle (Figure 6.4).

This is a variation on the objectives tree as described earlier and demonstrated in the other examples, and might more accurately be called a 'functions tree'. However, the same principles apply of breaking-down objectives into sub-objectives, or functions into means, and ordering them into a hierarchical tree. This application of the tree structure approach helps to ensure that all the possible means of achieving a function (or objective) are considered by the designer. (Source: Tjalve)

*Example 4:
Surgical shoe*

Surgical shoes or clogs are worn in hospital operating theatres. In order to promote aseptic procedures and maintain a sterile environment surgical staff wear special clothing, including shoes. In addition to the shoe's function of isolating users' feet from the sterile environment of surgical departments, designers are of course expected to provide for users' comfort and other relevant needs. In a study in Ghana, a market survey identified several problems associated with the commonly used surgical shoes. Among the problems were insole discomfort and some interviewees complained about the warmth generated at the vamp (central front) area of the shoes. The study also found a lack of availability of shoes in all foot sizes. Users of the surgical shoe were concerned about the performance, affordability, durability, appearance and safety of the product. An engineering design process was followed to investigate the problems and provide new solutions, including a shoe that could be adjusted to different sizes. To ensure that the needs of the end-users would be met, the initial objectives of the design were developed as shown in the objectives tree in Figure 6.5.

A size-adjustable surgical shoe was subsequently designed, employing a shock-absorbing polyimide insole to improve comfort.

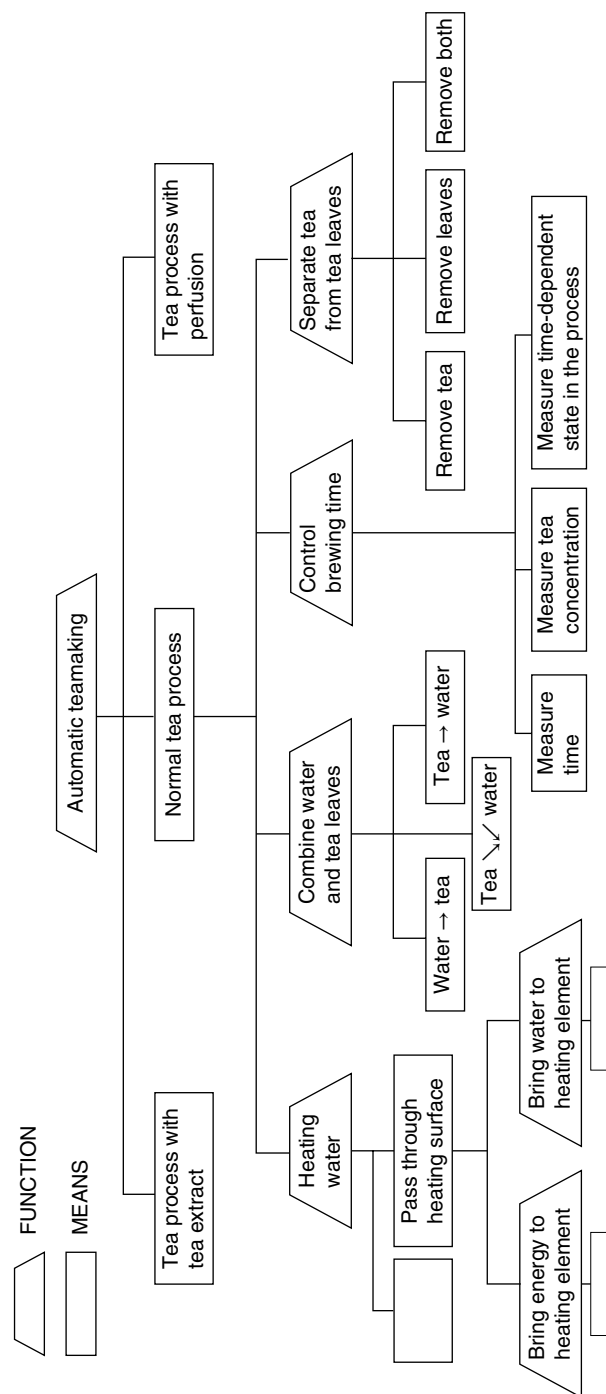


Figure 6.4 A functions-means tree for an automatic teamaker. Source: Tjalve, A Short Course in Industrial Design. Reproduced with permission of Newnes-Butterworth.

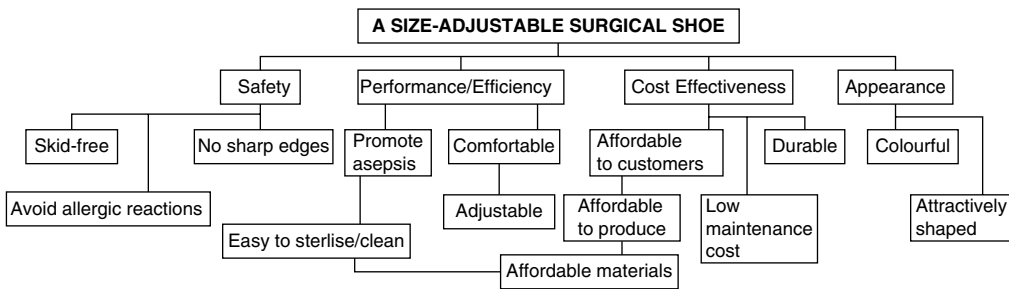


Figure 6.5 Objectives tree for a surgical shoe. Source: Atike & Kaufman, 4th International Conference on Appropriate Technology. Reproduced with permission.

It also had novel rack and pinion gear systems that could be adjusted three dimensionally to suit a wide range of foot sizes. Additionally, ventilation pores were provided to aerate the vamp area of the shoe while still maintaining sterile asepsis conditions. Figure 6.6 shows a concept design sketch developed for the shoe. (Source: Atike and Kaufmann)

Worked example:
High-pressure pump

This example is based on the design of a pump for high-pressure, high-temperature fluids. The manufacturers who commissioned the design already made a variety of such pumps, but wished to rationalise their range of pumps in order to reduce manufacturing costs.

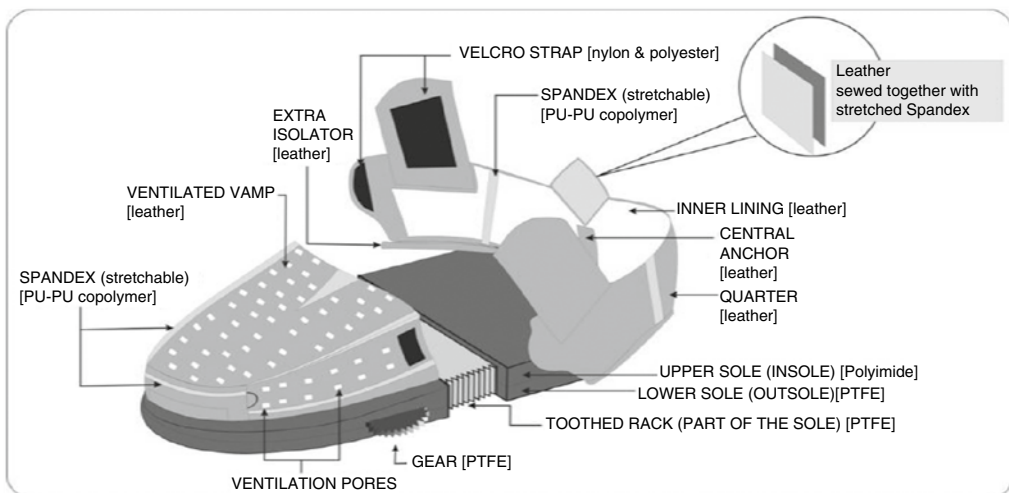


Figure 6.6 Concept drawing for the surgical shoe. Source: Atike & Kaufman, 4th International Conference on Appropriate Technology. Reproduced with permission.

They also wanted to improve the reliability of their pumps and to offer a product that was seen to be convenient to the varied and changing needs of their customers.

On questioning the client about the ‘reliability’ and ‘convenience’ objectives, a common aspect emerged: that the pump should be ‘robust’ – i.e. that it should not easily fail. The initial list of objectives so far might therefore look like this, in hierarchical order:

- reliable
- convenient
- robust
- standardised range

These are all still rather high-level and general objectives, so it is necessary to investigate such statements further. In this case, it was possible to investigate the problems experienced with the existing pumps. It was found that they were sometimes affected by cracking and leakages due to the stresses caused by the thermal expansion of the pipes to which they were connected. This appeared to be the main problem to which the requests for ‘robustness’ and ‘reliability’ were aimed.

Similarly, investigating the ‘convenient’ objective revealed a further two sub-objectives; firstly that the pumps should be easy to install and replace, and secondly that they should occupy the minimum space. It was realised that the standardisation of sizes and dimensions in the range could be a means of helping to achieve these objectives, as well as reducing manufacturing costs.

The expanded list of objectives therefore looked like this:

• Reliable	• Convenient
• Robust	• Easy to install and replace
• Resistant to external mechanical stresses	• Occupy minimum space
• Unaffected by thermal expansion of pipes	• Standardised range

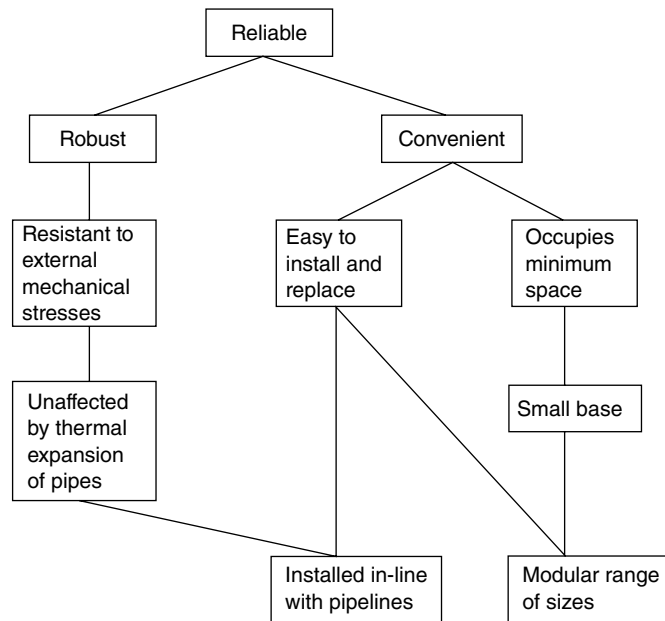


Figure 6.7 Objectives tree for a pump.

A key design principle to emerge from considering the means of achieving these objectives was that the inlet and outlet ports should always be in-line, to avoid the thermal expansion problems. Such a system, coupled with a small base size and modular dimensioning of alternative components, would also facilitate installation and replacement of the pump. The objectives tree therefore looked like Figure 6.7.

A high-pressure pump has been designed on similar principles in Denmark (see Figure 6.8). According to a description by the Danish Design Council, the pump is ‘almost a diagram of its problem statement: intake and discharge are aligned, motor, coupling and the stage-built pump are aligned on an axis at right angles to the installation surface, and the pump pressure is increased by adding to the number of stages, i.e. a change in height. The pump is installed directly on the pipeline, occupying a minimum of space.’



Figure 6.8 Grundfos pumps designed on similar principles to the objectives developed in this example. Source: Grundfos